OSAKPOLOR OGIEMUDIA (SURGE)

Full-Stack Product Engineer • Systems & Healthcare Infrastructure

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PROFESSIONAL SUMMARY

High-velocity **Full-Stack Product Engineer** architecting and shipping complex 0-to-1 software systems across offline retail, healthcare logistics, native desktop daemons, and client-side edge AI. Uniquely combines deep systems engineering (atomic multi-document ACID sessions, IndexedDB synchronization, WebSockets, WASM runtimes) with elite healthcare & regulatory domain authority (PharmD, MPH, WHO vaccine monitoring & NAFDAC regulatory experience). Proven record shipping 5 production systems.

TECHNICAL SKILLS

Languages & Core: TypeScript, JavaScript, Python, C++, HTML5/CSS3, SQL, Shell Scripting.
Full-Stack & Frameworks: Next.js (App Router), React, Node.js, Express, Electron, Tailwind CSS, REST APIs, GraphQL.
Databases & Offline Systems: MongoDB (Replica Sets, ACID Transactions), PostgreSQL, SQLite, Prisma ORM, IndexedDB, Redis.
AI, Vision & Edge: Pyodide (WebAssembly), WebGPU, Gemma 4, Claude 3.5 Sonnet, Gemini Multimodal Vision OCR.
Protocols & Hardware: WebSockets (Pusher), WhatsApp Webhook Daemons, ESC/POS Thermal Printing, HMAC-SHA512 Security.

PRODUCTION ENGINEERING & FLAGSHIP SYSTEMS

Lead Product Engineer & Architect • PharmaStackX Consumer Healthcare Network

2024 – Present

`Stack: Next.js 16, MongoDB (45 Schemas), Gemma 4, Pusher WebSockets, WhatsApp Cloud API, Paystack`

- Architected and shipped a nationwide multi-sided healthcare fulfillment engine across **2,000+ production commits**, connecting thousands of patients with licensed pharmacies in real time.
- Engineered an autonomous WhatsApp bot daemon running 24/7 in regional community groups, leveraging **Gemma 4** to extract unstructured pricing quotes into structured database records.
- Implemented multimodal vision OCR pipelines parsing handwritten prescription photos and pill packaging with automated error-correction.
- Built the "Ask Rx" clinical triage assistant with automated human-in-the-loop escalation protocols to on-call pharmacists.
- Designed dynamic multi-tenant subdomain routing (`slug.pharmastackx.com`) syncing pharmacy shelf inventories directly to patient ordering portals.

Full-Stack Systems Engineer • PSX Universal Point-of-Sale (Universal POS Engine)

2024 – Present

`Stack: Next.js App Router, IndexedDB, MongoDB Replica Sets, ACID Transactions, ESC/POS Hardware`

- Engineered a high-throughput, offline-first retail POS terminal across **1,100+ commits**, enabling zero-latency checkouts during power and internet outages via local client-side **IndexedDB**.
- Implemented idempotent transaction reconciliation using unique `offlineReceiptNumber` keys and **atomic MongoDB multi-document sessions**, preventing race conditions and double deductions on reconnect.
- Programmed a dynamic packaging hierarchy conversion engine (Carton → Box → Pack → Piece) with strict FIFO batch inventory depletion.
- Developed native **ESC/POS byte-buffer generation** for 58mm/80mm thermal receipt printers and hardware USB barcode scanning.

Lead Engineer • PSX Clinical EMR (Electronic Medical Records)

2024 – Present

`Stack: Next.js, Prisma ORM, MongoDB, Web Audio API, Gemini Clinical AI, Dispensary SSO`

- Built a standardized 6-stage outpatient clinical pipeline: Complaint → HPC → Vitals → Review of Systems → Assessment → Management Plan.
- Engineered client-side Web Audio consultation recording with real-time AI transcription and clinical diagnostic suggestion drafting.
- Implemented polymorphic tamper-evident audit logs capturing actor IDs, timestamps, and field-level diffs for medical record compliance.

Systems & Native Desktop Engineer • Synkk (Legacy POS-to-Cloud Sync Bridge)

2024 – Present

`Stack: Electron, Node.js, SQLite, Claude 3.5 Sonnet, Win32 Screen-Vision Hooks`

- Built a native Windows background daemon extracting live inventory from closed legacy desktop software with zero API access using screen vision and filesystem hooks.

- Built a Claude 3.5 Sonnet normalizer converting dirty, cryptic medicine names into standardized catalog records in real time.

Systems & AI Engineer • CS50P AI Tutor (Edge Compute Web IDE) 2024 – Present

`Stack: Next.js, WebAssembly (Pyodide), WebGPU, Gemma 4, Monaco Editor`

- Engineered an in-browser Python IDE running 100% on the user's client hardware with zero backend container hosting costs using **Pyodide WebAssembly**.
- Integrated on-device **WebGPU-accelerated Gemma 4** inference for instant, local syntax error diagnosis and code tutoring.

DOMAIN LEADERSHIP & CLINICAL EXPERIENCE

Vaccine Monitoring & Evaluation Consultant • World Health Organization (WHO) National Campaign

Led field data collection, logistics monitoring, and compliance evaluation for large-scale public health vaccination campaigns.

Public Relations Officer • Award-Winning Youth Leader • Young Pharmacists Group & PANS Leadership Tenure

Directed communications, organizational leadership, and stakeholder advocacy across regional healthcare professional bodies.

EDUCATION & CREDENTIALS

Master of Science (MSc) in Artificial Intelligence • UK / International Expected 2027

Master of Public Health (MPH) • Healthcare Policy & Systems Expected 2026

Doctor of Pharmacy (PharmD) • University of Benin 2024